

Blower Motor State Detection – FSM Documentation

Overview

This system monitors the AC current of a 3-speed blower motor and detects its operational state in real time.

States are:

- **OFF** – motor not running
- **LO** – low speed
- **MED** – medium speed
- **HI** – high speed

State detection is based on **RMS current measurements** sampled every 1 second.

The FSM (finite state machine) ensures accurate reporting while handling:

- Startup transients
- Short-duration partial window artifacts
- Clean timestamps for database logging

Measurement Setup

- **ADC:** ADS1115 (continuous mode)
 - **Channel:** AIN2
 - **Gain:** ± 2.048 V
 - **Sample rate:** 860 SPS
 - **RMS window:** 1 s
 - **Current conversion:** $I_{rms} = V_{rms} * 30$
 - **Startup discards:** first 3 RMS windows ignored
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State Thresholds

State RMS Current (A)

OFF < 4.5

LO 4.5 – 5.15

MED 5.15 – 5.6

HI > 5.6

Thresholds are programmable and can be tuned for specific motors.

FSM States

State	Description
INIT	Startup; determines initial motor state.
OFF	Motor not running. Monitors for transition to ON.
STARTUP_PENDING	Motor just exceeded OFF → LO threshold. Waits 10 s to account for high-current startup transient.
RUN_LO / RUN_MED / RUN_HI	Motor running at corresponding speed. Monitors for speed changes or OFF transitions with 2-second minimum dwell.

Transition Rules

1. INIT → OFF / RUN_*

- Determined by the first valid RMS window after startup settling.
- State is **reported immediately**.

2. OFF → STARTUP_PENDING

- Occurs when RMS current crosses LO threshold.
- Records the **epoch of first threshold crossing**.
- Enters **10 s startup delay** to ignore transient currents.

3. STARTUP_PENDING → RUN_LO / MED / HI

- After 10 s, RMS measurement classifies the new speed.
- **State is reported with timestamp from initial crossing**.
- Guarantees OFF → ON transitions are never misclassified.

4. RUN_* → RUN_* or OFF

- Only reported after **2 consecutive RMS windows** indicate the new state (minimum dwell).

- Timestamp corresponds to **first detection window**.
- Prevents spurious intermediate readings due to partial window artifacts.
- Examples:
 - MED → LO → LO confirmed after 2 windows → reported at first LO detection timestamp
 - LO → OFF → OFF confirmed after 2 windows → reported at first OFF detection timestamp

5. ON → OFF (motor stops)

- Minimum dwell of 2 seconds applies.
 - Prevents false OFF events due to RMS window edge effects.
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Reporting

- Output format (console / database-ready):

YYYY-MM-DD HH:MM:SS State=<OFF/LO/MED/HI> Irms=<current> A

- **Timestamp:**
 - For OFF → ON startup: epoch of first threshold crossing
 - For speed changes or ON → OFF: epoch of **first RMS window** that detected the new state
 - **Current:** RMS current at the reporting window (1 s after the first detection in dwell cases)
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Special Notes

1. Startup transient

- High current can last ~10 s.
- Handled via STARTUP_PENDING with delayed reporting.

2. Minimum dwell

- 2-second requirement ensures **partial RMS windows or momentary fluctuations** do not trigger spurious transitions.
- No extra hysteresis is required due to stable motor currents.

3. FSM Benefits

- Single sm_state variable simplifies logic.

- Eliminates scattered flags (startup_pending, deferred_report).
- Cleanly handles all OFF/ON and speed transitions.

4. Extensibility

- FSM can be easily extended for:
 - Fault states (overcurrent, sensor failure)
 - Logging additional diagnostics (time-in-state, counters)
 - Adjusting dwell or transient delays without impacting RMS measurement
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Example Output Transcript

```
2026-02-21 08:40:50 State=MED Irms=5.402 A
2026-02-21 08:47:47 State=LO Irms=5.009 A
2026-02-21 08:48:17 State=OFF Irms=0.003 A
2026-02-21 09:04:03 State=MED Irms=5.487 A
2026-02-21 09:13:56 State=LO Irms=4.984 A
2026-02-21 09:14:36 State=OFF Irms=0.003 A
```

- Note the **OFF events correspond to true motor stop**, and LO/MED events reflect stable speed readings, free of spurious transitions.
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This documentation captures **all current logic**, explains why **minimum dwell is sufficient**, and can be included in your project folder or wiki.